

Objective California Housing

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Dataset: California Housing 🏠

🎯 Our objective

We have information about different housing areas in California.

Our ML question is:

Can we use the characteristics of a housing area to predict its median house value?

So this is a supervised learning - regression problem because our target is a continuous numerical value.

Understanding our columns

Columns	Meaning	Role
Longitude	Geographic longitude of the area	Feature
latitude	Geographic latitude of the area	Feature
housing_median_age	Median age of houses in the area	Feature
total rooms	Total rooms in the area	Feature
total bedrooms	Total bedrooms in the area	Feature
population	Population of the area	Feature
households	Number of households	Feature
median income	Median income of the area	Feature
ocean_proximity	How close the area is to the ocean	Feature
median_house_value	Median house value	🎯 Target

Notice something important

Most features are numerical, but:

ocean_proximity

is categorical.

That's actually useful because later, when we reach encoding, we'll have a real categorical feature to preprocess.

🧠 Our ML objective
We're essentially asking:

longitude
latitude
house age
rooms
bedrooms
population
households
income
ocean proximity



ML algorithm



median house value

So:

X = housing information available to the model

y = house value we want the model to predict